

Akshay Dahiya

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EDUCATION

GEORGIA TECH

MS IN COMPUTER SCIENCE

Exp 2021 | Atlanta, USA

Specialization: Machine Learning

GPA: 4.0/4.0

Research: Counter-Plagiarism Tools

(Our research project **Jack Watson** got featured in **Forbes**, **TimesHigherEducation**, and multiple other places.)

DCRUST

B.TECH IN ELECTRONICS AND COMM.

May 2018 | Murthal, India

GPA: 7.05/10.0

SKILLS

Languages

Python, Javascript, C++ (basic), Haskell(basic)

Web

Django, Flask, React, D3.js, PostgreSQL, Redis, JSON-LD, Semantic Web, Web sockets

AI

Numpy, Pandas, Sklearn, OpenCV, Tensorflow, Pytorch, Fastai

Other

Algorithms, Data Structures, Linux, Docker, AWS, GCP, Inkscape, UX design

COURSEWORK

GEORGIA TECH

Artificial Intelligence

Machine Learning

Computer Vision

Reinforcement Learning

AI Ethics and Society

Knowledge-Based AI

Data and Visual Analytics

Information Security and Policies

Graduate Intro to Operating Systems

Graduate Algorithms (in progress)

UDACITY

Computer Vision Nanodegree

Fullstack Web Developer Nanodegree

React Developer Nanodegree

Frontend Web developer Nanodegree

Tech Entrepreneur Nanodegree

FAST.AI

Practical Data Ethics

Practical Deep Learning for Coders

LINKS

Homepage | Github | LinkedIn

Leetcode | Twitter

PROJECTS

LUNAR LANDER | OPENAI GYM ENV

November 2019

- Solved the Lunar Lander Gym environment achieving an average score of 202.96 over 100 episodes using Pytorch.
- Experimented with various Reinforcement learning techniques like DQN, Double DQN, Experience Replay, and Prioritized Experience Replay.
- Prepared a 5-page paper summarizing my experiments and analysis.

HYDRUS & HYDRA FLOCK DEMO | HYDRAECOSYSTEM

Aug 2017 – Present

- Hydrus** is a web framework built in Flask, that can create semantic web-based Restful APIs using a single specs file (Hydra Vocabulary)
- Built a simulation (**Hydra flock demo**) where multiple servers and clients interact autonomously to demonstrate the capabilities of semantic web.
- The project has 154 Stars and 123 Forks (to date) on Github.

EXPERIENCE

HYPERSONIX INC. | SOFTWARE DEVELOPER

March 2020 – Present | Bengaluru, India

- Designed and implemented the entire Authentication system
- Improved API performance by about 40% by optimizing the codebase and incorporating compression algorithms like Brotli
- Set up logging mechanism for platform codebase using ELK and Filebeat
- Developed multiple critical customer-facing components like Hints, Guided Tours and Learning Center

RESEARCH

GEORGIA TECH | JACK WATSON TEAM

Jan 2020 – Present

Worked under **Dr. Thad Starner** to develop Counter-Plagiarism tools.

Ongoing Research Projects -

- Jack Watson**: An AI agent that can monitor sites like Freelancer and catch students if they try to hire someone by posing as a freelancer.
- Examinator**: A tool to catch students who collaborate or cheat in take-home exams.
- JackMarker**: A tool to watermark code and detect plagiarism at scale.
- GitDown**: A tool to monitor Github repos and submit take down requests.

VOLUNTEER AND OTHER EXPERIENCE

2019-20 Org Admin | Google Summer of Code

2018 Graduate Programmer

2018 Org Admin | Google Summer of Code

2018 Tech Speaker

2017 Google Summer of Code Participant

2017-20 Mentor and Project Reviewer

HydraEcosystem.org

Fidelity International

Python HYDRA

Pycon, India

Python Software Foundation

Udacity

PUBLICATIONS

- [1] R. Apoorv, A. Dahiya, U. Sreeram, B. R. Patil, I. Irish, R. Graziano, and T. Starner. Examinator: A plagiarism detection tool for take-home exams. In *Proceedings of the Seventh ACM Conference on Learning @ Scale, L@S '20*, page 261–264, New York, NY, USA, 2020. Association for Computing Machinery.